

claude-windows / 8d4866bf-8edd-4da9-909e-022dde12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022dde12675\subagents\workflows\wf_3d035317-127\agent-a63f8c750765049d9.jsonl`  
- SHA-256: `9b4a2a689eb04cd7234bbdb037e1f40b9444e32253635910e38cd59c5aadbd`  
- Source modified: `2026-06-22T09:47:26+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_3d035317-127`  
- session_id: `8d4866bf-8edd-4da9-909e-022dde12675`
```

Transcript

1. user (2026-06-22T09:46:23.312Z)

You are fixing staleness in ONE doc of the KhelSutra repo, applying an audit that already ran. Be SURGICAL: minimal, targeted edits that match the audited fix; preserve the doc's voice, structure, and markdown formatting. Do NOT wholesale-rewrite a doc.

KhelSutra repo root: C:\Users\avidu\Projects\kheLSutra
ENGINE repo root (sibling, for cross-repo verification): C:\Users\avidu\Projects\badminton-highlight-indexer (origin/master freshly fetched)
Today: 2026-06-22. KhelSutra is on branch docs/freshness-sweep (off master).

YOUR DOC (edit ONLY this file): C:\Users\avidu\Projects\kheLSutra\docs\PORTABILITY.md

The audit findings are in this JSON file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022dde12675\tasks\w3dbzrpby.output
Shape: { result: { perDoc: [{doc, overallStatus, findings, summary}], staleFindings: [{doc, claim, docLocation, verdict, evidence, suggestedFix, severity}] } }.
Read it and select ONLY entries where doc === "docs/PORTABILITY.md". Your worklist = that doc's staleFindings. Also skim that doc's perDoc.findings for "unverifiable" items ? do NOT act on those unless you can now positively verify them.

For each stale finding:

1. RE-VERIFY it yourself against the LIVE repo before editing ? Read the doc lines + the cited code. For any claim about the ENGINE (paths like backend/main.py, /api/*, native_wasb_runner.py, auth/CORS), verify against the engine repo's origin/master: e.g. ``git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "<symbol>" origin/master -- backend/`` or ``git -C ... show origin/master:backend/main.py``. Trust nothing blindly ? the audit itself may be slightly off and engine line numbers DRIFT between PRs.
2. If confirmed, apply the suggestedFix as a SURGICAL Edit. For brittle ENGINE line-number anchors, prefer SYMBOL-based references (e.g. ``backend/main.py::get_config``) over chasing exact line numbers ? but only after confirming the symbol exists on engine origin/master.
3. DOCS ONLY: if a fix would require a CODE change (e.g. a comment in site/server.js), do NOT edit code ? record it under codeChangesNeeded.

Hard rules:

- Edit ONLY your assigned doc file. Run NO git WRITE commands (no add/commit/checkout/branch/stash) ? read-only git for verification only.
- Keep status markers HONEST: flip Todo->Done only WITH the PR number / file evidence; never invent a completion.
- If a finding is wrong, not reproducible, or too risky to auto-apply, SKIP it and explain under skipped.
- Do not "improve" beyond the findings; no scope creep.

Return the structured report.

2. assistant (2026-06-22T09:46:25.743Z)

I'll start by reading the audit findings and the target doc.

3. user (2026-06-22T09:46:28.394Z)

```
1      # Portability ? Cloudflare now, always-on VM if needed
2
3      > **Status:** design + proven (2026-06-20). **Decision:** keep the site/form host-
portable so
4      > moving off Cloudflare to an owner-run always-on VM ("if Cloudflare acts out") is an
**adapter
5      > swap, not a rewrite.**
6
7      ## The shape that makes it portable
8      - **Logic is runtime-agnostic.** `src/index.js` uses Web-standard `Request`/`Response`
and pure
9      helpers (`validate`, `toRow`, `isBot`, `handleRequest`) ? no host-specific calls.
10     - **All DB coupling is in adapters** (`src/store.js`): `d1Store` (Cloudflare D1) and
`sqliteStore`
11     (node:sqlite / better-sqlite3). `handleRequest(request, store)` takes the store by
injection.
12     - **One schema, one SQL.** `schema.sql` and the `INSERT` are plain SQLite, so D1 and a
VM's SQLite
13     file are byte-for-byte compatible ? zero dialect change. (Postgres later = minor
placeholder tweaks.)
14
15     ## Two hosts, same code
16     | | Cloudflare (now) | Always-on VM (fallback) |
17     |---|---|---|
18     | Entry | `src/index.js` (Worker `fetch`) | `server.js` (`node:http`) |
19     | Static | `env.ASSETS` | `server.js` serves `public/` |
20     | Storage | `d1Store(env.DB)` ? D1 | `sqliteStore(db)` ? `node:sqlite` file |
21     | Country | `cf-ipcountry` header | optional (GeoIP / null) |
22     | Deploy | git ? Workers Builds | `node server.js` under systemd |
23     | TLS/DNS | Cloudflare | keep Cloudflare in front, **or** Caddy/Let's Encrypt |
24
25     **Enforced, not asserted.** `node --test src/` runs BOTH paths and CI
(`.github/workflows/test.yml`,
26     Node 24) runs them on every PR, so the repo stays compatible with both hosts:
27     - `src/index.test.js` ? the Cloudflare Worker + D1 logic.
28     - `src/server.test.js` ? **boots the real `server.js`** (node:http + node:sqlite) and
round-trips it
29     over HTTP into SQLite (the same proof we run on the droplet, automated).
30
31     For a live check against a real VM: `SSH_TARGET=root@host scripts/droplet-smoke.sh` (scp
+ boot +
32     round-trip + cleanup). Verified live on the owner's droplet (Ubuntu 24.04, isolated
`/srv/khelsutra-site`).
33
34     ## Provider-portable provisioning ? destroy ? resume anywhere (scripted + TESTED)
35     Spinning up a fresh VM on **any** provider (DigitalOcean / GCE / any Ubuntu box) and
tearing it down
36     without losing data is a **first-class, scripted** flow ? so destroying a box and
resuming elsewhere is
37     routine, not risky.
38
39     - **`scripts/provision-vm.sh`** ? run from your LOCAL machine against a FRESH Ubuntu VM.
Pushes `site/`
40     via `git archive` (**private-repo-safe** ? no clone, token, or secret on the VM),
installs Node (?24,
41     so `node:sqlite` is unflagged), installs + starts a **systemd** service, opens the
firewall, and
42     **verifies** it serves (form + a real POST round-trip). Idempotent.
43     ...
```

```

44     SSH_TARGET=root@<ip> [SSH_KEY=~/.ssh/khelsutra_droplet] [PORT=8787]
[RESTORE_DB=./backup.db] \
45     site/scripts/provision-vm.sh
46     ```
47     - **`scripts/teardown-vm.sh`** ? **exports the SQLite state DB to a local timestamped
backup BEFORE**
48     anything is destroyed (registrations survive the move), stops the service, and
optionally destroys the
49     DO droplet (`DESTROY=1 DROPLET_ID=<id>`, via `doctl`).
50     ```
51     SSH_TARGET=root@<ip> [BACKUP_DIR=./vm-state-backups] [DESTROY=1 DROPLET_ID=<id>] \
52     site/scripts/teardown-vm.sh
53     ```
54     - **Clean resume on a different provider:** `teardown-vm.sh` (export) ? destroy ?
`RESTORE_DB=<backup>`
55     SSH_TARGET=root@<newip> site/scripts/provision-vm.sh`. The restored DB is `scp`'d in
before the service starts.
56
57     **Verified live (2026-06-20):** a fresh DO Ubuntu-24.04 droplet went bare ? operational
+ publicly
58     serving (HTTP 200) via `provision-vm.sh` (Node 24 auto-installed, systemd unit,
firewall); `teardown-vm.sh`
59     exported the state DB (12 KB); a re-provision with `RESTORE_DB` preserved the
registration (`rows:2` =
60     restored + the verify POST) ? then the throwaway droplet was destroyed. The
destroy?resume cycle is
61     reproducible across providers.
62
63     ## VM deploy runbook ("if Cloudflare acts out")
64     Node 22+ only ? **no `npm install`** (node:sqlite is built in).
65     ```
66     # on the VM:
67     git clone ? && cd khelsutra/site          # or scp the site/ folder
68     PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
69     # put it under systemd (auto-restart) + a reverse proxy (Caddy/nginx) for TLS,
70     # then point DNS at the VM. ~1 hour; logic + schema unchanged.
71     ```
72
73     ## Honest notes
74     - "Cloudflare acts out" usually means **Workers/D1 pricing or limits, not DNS** ? so the
cheapest
75     move is to keep Cloudflare DNS/CDN in front and shift only the **compute** to the VM
(point a
76     route/origin at the box). Full exit (own DNS + Caddy TLS) is also supported.
77     - You can run **both** (Cloudflare primary + VM warm standby) since the schema is
identical; a
78     periodic `access_requests` export keeps them in sync.
79     - `server.js` is intentionally dependency-free; if you later want Postgres or a
framework (Hono runs
80     on Workers *and* Node), it slots in behind the same `store` + handler.
81

```

4. user (2026-06-22T09:46:28.572Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-591 of 1722 total (61859 tokens, cap 25000). Call Read with offset=592 limit=591 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1      {
2      "summary": "Audit every KhelSutra doc for staleness against the live repo, returning
structured findings",
3      "agentCount": 15,
4      "logs": [
5      "audited 15/15 docs; 44 stale findings; archive-candidates: none"

```

```

6      ],
7      "result": {
8          "perDoc": [
9              {
10                 "doc": "NEXT_STEPS.md",
11                 "overallStatus": "minor-staleness",
12                 "summary": "NEXT_STEPS.md is overwhelmingly accurate against master: routes,
file paths, the Worker/D1 names, the 6-language localization, the 3-step wizard, the per-rally
PR numbers (#11/#14/#21/#28/#30/#31), the appliance/per-rally doc statuses, and every still-
pending \"Queued\" item all check out against code and git. The only real staleness is the \"PRs
through #93 merged\" count, which lags by one because PR #94 (the very commit that wrote
\"through #93\") merged afterward; plus a low-severity GCS-artifact filename mismatch. Not an
archive candidate ? it is the live next-steps tracker.",
13                 "findings": [
14                     {
15                         "claim": "PRs through #93 merged (master).",
16                         "docLocation": "NEXT_STEPS.md:13",
17                         "verdict": "stale",
18                         "evidence": "git log --oneline -1 = '50b2bd7 chore(repo): ... correct stale
NEXT_STEPS claims (#94)'; total 94 commits/PRs on master. Ironically #94 is the commit that
updated this line to '#93', then merged itself, so the doc now lags one PR behind master.",
19                         "suggestedFix": "Update to 'PRs through #94 merged (master).'",
20                         "severity": "medium"
21                     },
22                     {
23                         "claim": "Where we are (2026-06-20): ... khelsutra.guru is LIVE and fully
shipped (landing + /capture checklist + OG image + segmented Request-Access form).",
24                         "docLocation": "NEXT_STEPS.md:5-14",
25                         "verdict": "correct",
26                         "evidence": "site/server.js:98 maps /capture->capture.html;
site/public/og.png + capture.html tracked; index.html:191-193 textareas + lines 537-581
academy/individual/gpu/storage fields. Header date 2026-06-20 is slightly behind the latest work
(PR #94 merged 2026-06-22) but the substance holds.",
27                         "suggestedFix": "Optionally bump the 'Where we are' date to 2026-06-22 to
match the latest merged work.",
28                         "severity": "low"
29                     },
30                     {
31                         "claim": "On Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-
access): the form persists to D1 and emails a summary via an Email Routing send binding.",
32                         "docLocation": "NEXT_STEPS.md:7-9",
33                         "verdict": "correct",
34                         "evidence": "site/wrangler.toml: name = \"hidden-king-khelsutra\";
[[d1_databases]] database_name = \"khelsutra-access\"; [[send_email]] name = \"NOTIFY\".
site/src/index.js:80 handles POST /api/request-access -> insert into D1 binding DB.",
35                         "suggestedFix": "",
36                         "severity": "low"
37                     },
38                     {
39                         "claim": "CI gates every PR ? three jobs: full site npm test, web/ SPA build
+ coverage floor, and a real-browser layout check.",
40                         "docLocation": "NEXT_STEPS.md:9-12",
41                         "verdict": "correct",
42                         "evidence": "site/package.json scripts include test (node --test),
test:layout (layout.browser.mjs); web has a coverage floor (#80). PR #94 message confirms 'three
jobs: site npm test = 86 green, web SPA build+coverage, real-browser layout'. Non-browser site
suite ran green (61 tests in the 4 non-layout files).",
43                         "suggestedFix": "",
44                         "severity": "low"
45                     },
46                     {
47                         "claim": "The form also runs on a dependency-free Node + SQLite host
(site/server.js), live on the D0 droplet -> http://168.144.126.176:8787 (systemd khelsutra-
site).",
48                         "docLocation": "NEXT_STEPS.md:11-13",

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49         "verdict": "correct",
50         "evidence": "site/server.js exists, uses node:sqlite (header comment line
2). IP 168.144.126.176 + port 8787 + 'khelsutra-site' all appear in
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48 ('/srv/khelsutra-site, node :8787') and deploy refs.
Live-status is external/unverifiable from the repo, but the host code and identifiers are
correct.",
51         "suggestedFix": "",
52         "severity": "low"
53     },
54     {
55         "claim": "Docs: docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md.",
56         "docLocation": "NEXT_STEPS.md:13-14",
57         "verdict": "correct",
58         "evidence": "git ls-files: both docs/REQUEST_ACCESS_FORM.md and
docs/PORTABILITY.md are tracked.",
59         "suggestedFix": "",
60         "severity": "low"
61     },
62     {
63         "claim": "DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22): localized to
all 6 languages (en/kn/hi/es/da/id); full record docs/archives/FRIENDLY_AND_MULTILINGUAL.md.",
64         "docLocation": "NEXT_STEPS.md:17-21",
65         "verdict": "correct",
66         "evidence": "docs/archives/FRIENDLY_AND_MULTILINGUAL.md is tracked; git log
shows PR #93 'docs: archive the Friendly & Multilingual tracker'. web/src/i18n/index.ts:25
SUPPORTED_LOCALES = [en,kn,hi,es,da,id]; PR #87 added es/da/id to the marketing site.",
67         "suggestedFix": "",
68         "severity": "low"
69     },
70     {
71         "claim": "Interactive per-rally selection MVP/M0 COMPLETE; Done: P0 design
(#11), P1 scaffold (#14), P2 grid+hook (#21), P3 parseQuery query bar (#28), P4 SelectionFooter
(#30), P5 real load+download (#31). Tracked doc docs/PER_RALLY_SELECTION.md (IN PROGRESS).",
72         "docLocation": "NEXT_STEPS.md:32-41",
73         "verdict": "correct",
74         "evidence": "git log: cb74ce5 #11 'tracked design ... RallyPicker'; aac7e0c
#14 'P1 scaffold'; 62598e7 #21 'P2 selectable rally grid + hook'; d537d19 #28 'P3 deterministic
parseQuery() query bar'; aff8de8 #30 'P4 SelectionFooter + min_shot_count warning'; 8bf200d #31
'P5 load real rallies + Build->compile->preview->download'. docs/PER_RALLY_SELECTION.md status
header = 'IN PROGRESS ? MVP/M0 (P1-P5) COMPLETE'.",
75         "suggestedFix": "",
76         "severity": "low"
77     },
78     {
79         "claim": "P2c first-run setup wizard ? 3-step SetupWizard from
/api/capabilities, 6 languages.",
80         "docLocation": "NEXT_STEPS.md (implied via track) and verified for the
merged #88",
81         "verdict": "correct",
82         "evidence": "web/src/components/SetupWizard.tsx:22 'const STEPS = 3'; line 1
comment 'Three quick steps'; line 3 'Localized (en/kn/hi/es/da/id)'; web/src/api/client.ts:139
GET /capabilities. git: 7ae9cee #88.",
83         "suggestedFix": "",
84         "severity": "low"
85     },
86     {
87         "claim": "Single-box appliance ? Tracked design
docs/LOCAL_APPLIANCE_ARCHITECTURE.md (DRAFT, owner-gated).",
88         "docLocation": "NEXT_STEPS.md:52-58",
89         "verdict": "correct",
90         "evidence": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md status header: 'Status:
DRAFT ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge)'.",
91         "suggestedFix": "",
92         "severity": "low"
93     },

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94         {
95             "claim": "Queued: Always-Use-HTTPS + HSTS (Cloudflare) and Cloudflare Web
Analytics token paste into site/public/index.html ? both still pending.",
96             "docLocation": "NEXT_STEPS.md:44-45",
97             "verdict": "correct",
98             "evidence": "No Strict-Transport-Security/HSTS header found in site/*.js or
wrangler.toml. site/public/index.html:629 still has data-cf-beacon token 'REPLACE_WITH_CF_TOKEN'
(commented-out block). Both correctly listed as not-yet-done.",
99             "suggestedFix": "",
100            "severity": "low"
101        },
102        {
103            "claim": "Droplet -> production: add VM-path SMTP so the Node host also
emails (currently noopNotify).",
104            "docLocation": "NEXT_STEPS.md:46-47",
105            "verdict": "correct",
106            "evidence": "site/server.js:14 imports noopNotify; line 24 'const notify =
noopNotify(); // VM host: no email yet (add SMTP later)'.",
107            "suggestedFix": "",
108            "severity": "low"
109        },
110        {
111            "claim": "Org consolidation DONE; khelsutra (this site repo) transfer is an
owner step still PENDING (unchecked).",
112            "docLocation": "NEXT_STEPS.md:65-73",
113            "verdict": "correct",
114            "evidence": "git remote -v shows origin =
https://github.com/avidullu/khelsutra.git (NOT under Khelsutra org) ? consistent with the '?
owner step pending' marker. Other repos already moved: docs/DRY_RUN_CUJ.md:49 links
github.com/Khelsutra/badminton-highlight-indexer#289.",
115            "suggestedFix": "",
116            "severity": "low"
117        },
118        {
119            "claim": "A real-UI CUJ screencast exists -> promo_cuj.mp4/.gif in
gs://khelsutra-rally-corpus/promo/.",
120            "docLocation": "NEXT_STEPS.md:29-31",
121            "verdict": "unverifiable",
122            "evidence": "No GCS access to confirm the object. The companion doc
docs/DRY_RUN_CUJ.md:85 names the actual artifacts 'khelsutra-bucket-cuj-screencast.mp4' and ':60
khelsutra-i18n-screencast.mp4' in the same gs://khelsutra-rally-corpus/promo/ prefix;
'promo_cuj.mp4' appears nowhere else in the repo, suggesting a possible filename mismatch.",
123            "suggestedFix": "Reconcile the filename with docs/DRY_RUN_CUJ.md (likely
'khelsutra-bucket-cuj-screencast.mp4') or confirm 'promo_cuj.mp4' is a real object in the
bucket.",
124            "severity": "low"
125        },
126        {
127            "claim": "Control plane (control-plane/) at beta.",
128            "docLocation": "NEXT_STEPS.md:62",
129            "verdict": "correct",
130            "evidence": "control-plane/ directory does not exist in the repo (test -d
failed), but this line sits under 'Bigger arcs (design-first)', i.e. a forward-
looking/aspirational item, not a claim of present existence ? so it is not stale as written.",
131            "suggestedFix": "",
132            "severity": "low"
133        }
134    ]
135 },
136 {
137     "doc": "README.md",
138     "overallStatus": "significant-staleness",
139     "summary": "The repo-structure list, deploy references, and canonical-doc
pointers are all correct, but the README's headline status is badly out of date: it still calls
the repo a \"pre-alpha scaffold\" whose SPA has yet to land \"per milestone M1\", whereas the

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web/ SPA is in fact built, multilingual, and shipped (per-rally MVP P0?P5, P2c wizard, in-UI
upload, 6 languages; khelsutra.guru is LIVE). The control-plane/ directory it lists also does
not yet exist.",
140         "findings": [
141             {
142                 "claim": "Status: pre-alpha scaffold. SPA lands in `web/` per execution-plan
milestone M1.",
143                 "docLocation": "README.md:9",
144                 "verdict": "stale",
145                 "evidence": "web/src/ has 51 tracked source files incl. web/src/App.tsx;
`git log --oneline -- web/` = 31 commits; merged PRs show the SPA is feature-complete: P2c
first-run wizard (#88), in-UI file upload T4 (#66), per-rally selection MVP P0?P5
(#11/#14/#21/#28/#30/#31), 6-language catalogs (#67). NEXT_STEPS.md:7 states 'khelsutra.guru is
LIVE and fully shipped'. UI_ALPHA_EXECUTION_PLAN.md:36 defines M1 only as the initial
'Vite+React+TS+Tailwind scaffold', which is long past.",
146                 "suggestedFix": "Replace with a current status, e.g.: 'Status: alpha ? the
web/ SPA is built and multilingual (6 languages), with the per-rally selection MVP and first-run
setup wizard shipped; khelsutra.guru / app.khelsutra.guru are live (Cloudflare Pages +
cloudflared tunnel + Access). See NEXT_STEPS.md.'",
147                 "severity": "high"
148             },
149             {
150                 "claim": "the web app (alpha SPA) now, the hosted control plane at beta",
151                 "docLocation": "README.md:4",
152                 "verdict": "stale",
153                 "evidence": "This tagline pairs with line 9's 'pre-alpha scaffold', which is
contradicted by the shipped, live SPA (see web/ evidence above and NEXT_STEPS.md:7). The 'alpha
SPA now' framing is directionally right but undercut by the stale 'pre-alpha scaffold' status it
sits beside.",
154                 "suggestedFix": "Keep 'web app (alpha SPA) now' but align the rest of the
intro/status so it no longer describes the SPA as not-yet-landed.",
155                 "severity": "medium"
156             },
157             {
158                 "claim": "`control-plane/` ? (beta) Cloud Run FastAPI control plane",
159                 "docLocation": "README.md:17",
160                 "verdict": "stale",
161                 "evidence": "No control-plane/ directory exists: `ls control-plane` -> 'No
such file or directory'; `git ls-files control-plane/` returns nothing. Top-level tracked dirs
are deploy/, docs/, locales/, site/, web/ only.",
162                 "suggestedFix": "Mark it as a future/planned directory that does not yet
exist (e.g. move it out of the 'Structure' list into a 'Planned (beta)' note), so a reader
doesn't expect the path to be present.",
163                 "severity": "medium"
164             },
165             {
166                 "claim": "Engine ... sibling repo `badminton-highlight-indexer` (rename
pending).",
167                 "docLocation": "README.md:6",
168                 "verdict": "correct",
169                 "evidence": "Sibling repo still exists under that name: `ls
C:/Users/avidu/Projects/` shows 'badminton-highlight-indexer'; no rename has occurred, so
'rename pending' holds.",
170                 "suggestedFix": "",
171                 "severity": "low"
172             },
173             {
174                 "claim": "Canonical planning docs live in the engine repo's docs/ for now:
UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md.",
175                 "docLocation": "README.md:7-8",
176                 "verdict": "correct",
177                 "evidence": "All three exist in the engine repo (badminton-highlight-
indexer/docs/UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md present) and are NOT
tracked in khelsutra/docs (`git ls-files docs/` in khelsutra returns none of them; `ls
docs/UI_PROPOSAL.md` -> not found).",

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```

178         "suggestedFix": "",
179         "severity": "low"
180     },
181     {
182         "claim": "`deploy/` ... alpha deploy kit ... DEPLOY_ALPHA.md ... +
cloudflared.config.example.yml",
183         "docLocation": "README.md:15",
184         "verdict": "correct",
185         "evidence": "git ls-files deploy/ shows deploy/DEPLOY_ALPHA.md and
deploy/cloudflared.config.example.yml (plus deploy/droplet/*). DEPLOY_ALPHA.md:1-5 matches the
described 'shareable page, no SSH; SPA on Pages + engine on api.* tunnel + CF Access' content.",
186         "suggestedFix": "",
187         "severity": "low"
188     },
189     {
190         "claim": "`site/` ? public marketing landing page (static; served at
khelsutra.guru)",
191         "docLocation": "README.md:13",
192         "verdict": "correct",
193         "evidence": "site/ exists with site/public/index.html, server.js, etc.;
DEPLOY_ALPHA.md and NEXT_STEPS.md confirm the marketing site is served at khelsutra.guru (apex)
while the SPA is at app.khelsutra.guru.",
194         "suggestedFix": "",
195         "severity": "low"
196     },
197     {
198         "claim": "`web/` ? React SPA (alpha; Vite + React + TS + Tailwind;
Cloudflare Pages build root)",
199         "docLocation": "README.md:14",
200         "verdict": "correct",
201         "evidence": "web/ is a Vite+React+TS+Tailwind SPA (web/README.md:3 'Vite +
React + TS + Tailwind v4'); DEPLOY_ALPHA.md:14 confirms 'app.khelsutra.guru -> Cloudflare Pages
(the web/ SPA)'. The tech-stack/build-root facts are accurate; only the inline 'alpha' maturity
ties into the stale status above.",
202         "suggestedFix": "",
203         "severity": "low"
204     },
205     {
206         "claim": "License: All rights reserved (private; no OSS license).",
207         "docLocation": "README.md:19-21",
208         "verdict": "correct",
209         "evidence": "No LICENSE file is tracked (not present in `git ls-files`);
consistent with a private 'all rights reserved' repo.",
210         "suggestedFix": "",
211         "severity": "low"
212     }
213 ]
214 },
215 {
216     "doc": "deploy/DEPLOY_ALPHA.md",
217     "overallStatus": "minor-staleness",
218     "summary": "Verified correct except a stale header",
219     "findings": [
220         {
221             "claim": "header stale deploy live",
222             "docLocation": "deploy/DEPLOY_ALPHA.md:3",
223             "verdict": "stale",
224             "evidence": "droplet/README.md:100 live-verified 2026-06-22",
225             "suggestedFix": "set header to LIVE",
226             "severity": "medium"
227         },
228         {
229             "claim": "host list too narrow",
230             "docLocation": "deploy/DEPLOY_ALPHA.md:66",
231             "verdict": "stale",

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232         "evidence": "engine.env.example adds loopback",
233         "suggestedFix": "add localhost",
234         "severity": "low"
235     }
236 ]
237 },
238 {
239     "doc": "deploy/droplet/README.md",
240     "overallStatus": "fresh",
241     "summary": "This command-dense droplet quickstart is accurate and matches the
live repo: every cited kit file exists with the described contents, and the engine-behavior
claims (fail-fast boot posture, RALLY_* exposure vars, the Gemini-default-on-CPU gotcha, the
gs:// jail rejection, the --server entrypoint) all verify against the engine repo and
DEPLOY_ALPHA.md. The only items not locally reproducible are the 2026-06-22 droplet live-run
figures (e.g. \"1313 passed\"), which are an off-box assertion ? they are internally plausible
(engine has ~1269 test functions before parametrization) but cannot be confirmed from these
repos.",
242     "findings": [
243     {
244         "claim": "Four kit files exist: config.json.example, engine.env.example,
khelsutra-engine.service (and ../cloudflared.config.example.yml), each going to the documented
destination with the documented purpose.",
245         "docLocation": "deploy/droplet/README.md:9-14",
246         "verdict": "correct",
247         "evidence": "git ls-files deploy/ lists all four:
deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service} and
deploy/cloudflared.config.example.yml; contents read and match (config.json.example has the
security block + cf_team/aud; engine.env.example has the two RALLY_* exposure vars +
RALLY_APP_HOME; the service binds 127.0.0.1:8000; cloudflared.config routes api. ->
127.0.0.1:8000).",
248         "suggestedFix": "",
249         "severity": "low"
250     },
251     {
252         "claim": "The boot fails fast (exit 1, '[STARTUP] refusing to start') if
edge_auth_enabled=true but the jail or cf_team_domain/cf_access_aud is unset.",
253         "docLocation": "deploy/droplet/README.md:53-54",
254         "verdict": "correct",
255         "evidence": "backend/main.py:857 logs '[STARTUP] refusing to start';
backend/config/startup.py:109 EXPOSED_NO_VIDEO_ROOT and :123 EXPOSED_EDGE_AUTH_INCOMPLETE are
both LEVEL_ERROR (startup.py:41,272-276), while EXPOSED_AUTH_EXTRA_MISSING (PyJWT) is LEVEL_WARN
(:135). From engine PR #294 (commit 0c0e064).",
256         "suggestedFix": "",
257         "severity": "low"
258     },
259     {
260         "claim": "indexing.default_segmenter MUST be gemini on a CPU box; the
shipped default yolo_hybrid needs torch/GPU and every job fails 'No module named torch' without
it.",
261         "docLocation": "deploy/droplet/README.md:109-113",
262         "verdict": "correct",
263         "evidence": "config.json.example:9-11 sets default_segmenter=gemini. Engine
AppConfig default is yolo_hybrid (backend/config/models.py:128), so an all-defaults load on the
droplet (which hand-writes config.json rather than running the --app LIGHT seeder) does land on
yolo_hybrid. The LIGHT 'motion' default (loader.py:227) only applies via
write_light_default_config (indexer --app first-run, PR #310), not this systemd path ? so the
gotcha is accurate for this deploy.",
264         "suggestedFix": "",
265         "severity": "low"
266     },
267     {
268         "claim": "RALLY_ALLOWED_HOSTS is the load-bearing fix; with edge_auth on,
the Host-header guard defaults loopback-only and 400s every tunneled request until set.",
269         "docLocation": "deploy/droplet/README.md:92, engine.env.example:14-17",
270         "verdict": "correct",

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271         "evidence": "backend/config/startup.py:143-147 raises
EXPOSED_HOST_ALLOWLIST_LOOPBACK (LEVEL_WARN) when edge_auth is ON and RALLY_ALLOWED_HOSTS is
unset, with the exact 'will 400 forwarded requests' rationale; backend/main.py reads
RALLY_ALLOWED_HOSTS / RALLY_BIND_HOST / RALLY_CORS_ALLOW_ORIGINS. Matches DEPLOY_ALPHA.md §2.2
repro (400->200).",
272         "suggestedFix": "",
273         "severity": "low"
274     },
275     {
276         "claim": "Pasting a gs:// URI is rejected by the jail on the exposed box;
upload is the alpha path.",
277         "docLocation": "deploy/droplet/README.md:96-98",
278         "verdict": "correct",
279         "evidence": "tests/test_api_endpoints.py:608
test_process_hosted_mode_rejects_nonlocal_uri pins the gs:// -> 400 behavior under
allowed_video_root (referenced at :541 as the counterweight); the jail is required to expose
(config.json.example sets allowed_video_root).",
280         "suggestedFix": "",
281         "severity": "low"
282     },
283     {
284         "claim": "Pages build: Root dir web, Build npm run build, Output web/dist,
with VITE_API_BASE = https://api.khelsutra.guru/api.",
285         "docLocation": "deploy/droplet/README.md:73-76",
286         "verdict": "correct",
287         "evidence": "khelsutra web/ SPA exists (git ls-files web/...);
web/package.json build script = 'vite build'; web/src/api/client.ts:25 reads
import meta.env.VITE_API_BASE (default '/api'), and client.test.ts:32 exercises the exact split
value https://api.khelsutra.guru/api. Vite default outDir 'dist' relative to root web/ =
web/dist.",
288         "suggestedFix": "",
289         "severity": "low"
290     },
291     {
292         "claim": "The shipped engine is run via the auth extra and python -m
backend.main --server, binding 127.0.0.1:8000.",
293         "docLocation": "deploy/droplet/README.md:33,47-49; khelsutra-
engine.service:26",
294         "verdict": "correct",
295         "evidence": "backend/main.py:867 defines the --server flag and :1107 calls
uvicorn.run(app, host=bind_host, port=port); --setup flag at :865; the service ExecStart uses
python -m backend.main --server.",
296         "suggestedFix": "",
297         "severity": "low"
298     },
299     {
300         "claim": "Live-verified on the droplet (2026-06-22): full suite 1313 passed
+ ruff/mypy clean on the box; boot-posture, Host-guard 400->200, path-jail, T4 upload->jail,
edge-auth 401-gate exercised live; full DNS->Cloudflare->Access->tunnel->engine chain
confirmed.",
301         "docLocation": "deploy/droplet/README.md:100-106",
302         "verdict": "unverifiable",
303         "evidence": "This is an off-box live-run assertion that cannot be reproduced
from the khelsutra or engine repos here. Sanity check: the engine tests/ dir has ~1269 'def
test_' functions (parametrize expansions push the collected/passed count higher), so '1313
passed' is plausible and consistent, not contradicted. Marked unverifiable per the file:line-
evidence discipline; no sign of staleness.",
304         "suggestedFix": "",
305         "severity": "low"
306     },
307     {
308         "claim": "Cost: Cloudflare Pages + Tunnel + Access are free (Access <= 50
users); standing cost ~ $0 + Gemini per run.",
309         "docLocation": "deploy/droplet/README.md:16-18",
310         "verdict": "unverifiable",

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311         "evidence": "External Cloudflare pricing claim; not checkable against the
repo. Consistent with DEPLOY_ALPHA.md:3 ('~$0/mo standing ... + Gemini per-run'). No internal
contradiction.",
312         "suggestedFix": "",
313         "severity": "low"
314     }
315 ]
316 },
317 {
318     "doc": "docs/ANNOTATE_AND_TRAIN.md",
319     "overallStatus": "significant-staleness",
320     "summary": "A DRAFT tracker whose marketing-page deliverables (P0-P4 plus the
real annotator screenshots P5 mentions) are all shipped and merged (#15/#22/#23/#25), yet the
doc was last touched 2026-06-20 (#23) and never updated to reflect that. The core
engine/annotator technical claims (schema bridge, ship-gate refusing a commercial ship, MIT
annotator v1.6.4, guardrails) all still verify as correct ? only their file:line citations have
drifted. Main staleness: status still DRAFT despite shipped work, P5 marked Todo though
annotator screenshots are committed, the doc is NOT actually indexed in docs/README.md (claimed
three times), and the page since grew 6-language localization the tracker never anticipated. It
is an archive candidate once honestly closed out.",
321     "findings": [
322     {
323         "claim": "Status: DRAFT ? owner-gated on scope (header, line 3); Last
updated: 2026-06-20.",
324         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:3",
325         "verdict": "stale",
326         "evidence": "git log --follow shows the marketing-page work merged: P0 doc
(#15, 6b62ebf), own-model page P1-P4 (#22, b1110e1), org link (#23, 87520b8). Per §7, P0-P4 are
all marked Done ? yet the header still reads DRAFT. Lifecycle (line 4) is DRAFT->IN
PROGRESS->DONE; with P0-P4 shipped this is at least IN PROGRESS.",
327         "suggestedFix": "Flip Status to IN PROGRESS (only owner-gated P5-P9 remain)
and bump Last updated; or, if the owner considers the marketing surface complete, mark DONE and
archive per the lifecycle note.",
328         "severity": "medium"
329     },
330     {
331         "claim": "P5 'Capture + commit real screenshots ... replace placeholders' ?
Status ? (Todo), Gated Owner.",
332         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:76",
333         "verdict": "stale",
334         "evidence": "PR #25 (1906c82) 'show the real annotator screenshots'
committed site/public/img/annotate-enable.png (133712 bytes) and annotate-tool.png (153133
bytes); own-model.html:111-112 references them as the real VLC-UI shots (no people/footage). The
annotator-UI half of P5 is done and merged, not Todo.",
335         "suggestedFix": "Mark P5 ? for the annotator-UI screenshots (cite #25); if
footage screenshots with consented people are still pending, split that into its own row so the
done part isn't hidden under a blanket Todo.",
336         "severity": "high"
337     },
338     {
339         "claim": "Doc is 'index[ed] in docs/README.md' (line 5), §7 P0 'indexed in
docs/README.md' (? , #15), and DoD 'doc indexed in docs/README.md'.",
340         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:5,71,91",
341         "verdict": "stale",
342         "evidence": "grep -ni 'annotate|own.model|train.your|ANNOTATE_AND'
docs/README.md returns NOT FOUND; the docs/README.md 'Tracked work' list (lines 1-30) does not
mention ANNOTATE_AND_TRAIN.md or own-model. The claimed index entry does not exist.",
343         "suggestedFix": "Either add an ANNOTATE_AND_TRAIN.md entry to
docs/README.md's 'Tracked work' list, or correct P0/DoD to reflect that it is not yet indexed.",
344         "severity": "high"
345     },
346     {
347         "claim": "Engine ship-gate (gate_verdict, harness.py:98) currently refuses a
commercial ship regardless of F1: WASB provenance C3 unresolved + F1@tIoU not calibrated
(#172).",

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348         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:12,31",
349         "verdict": "correct",
350         "evidence": "training/gen0/harness.py:177 returns {'ship': False, ...}; :175
'WASB checkpoint provenance C3 UNRESOLVED ? commercial ship blocked regardless of F1'; :174
'ship-gate F1@tIoU target NOT YET CALIBRATED ... see issue #172'. Fact correct; the cited line
(:98) has drifted ? gate_verdict is now at :142.",
351         "suggestedFix": "Update the line citations: gate_verdict is at
harness.py:142 (not :98).",
352         "severity": "low"
353     },
354     {
355         "claim": "Default synth path: synth_trajectory_from_labels uses
csv.DictReader reading row['start_time']/row['end_time'] (stub_runner.py:139) ? MATCHES the
annotator header exactly.",
356         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:46",
357         "verdict": "correct",
358         "evidence": "backend/pipeline/detectors/stub_runner.py:124 def
synth_trajectory_from_labels; :137 for row in _csv.DictReader(f); :139
rallies.append((float(row['start_time']), float(row['end_time']))). Annotator header
rally_number,start_time,end_time,... confirmed at rally-annotator vlc/rally_annotator.lua:128
and docs/CSV_FORMAT.md:6. Claim accurate.",
359         "suggestedFix": "",
360         "severity": "low"
361     },
362     {
363         "claim": "Canonical GT reader load_gt_intervals accepts start/end OR
start_time/end_time (gt_loader.py:14-90).",
364         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:47",
365         "verdict": "correct",
366         "evidence": "backend/eval/gt_loader.py:30 _START_NAMES = ('start',
'start_time'); :44 def load_gt_intervals; :70 headerless positional col0,col1 path. Both aliases
accepted as claimed.",
367         "suggestedFix": "",
368         "severity": "low"
369     },
370     {
371         "claim": "Bundle attests no_paid_llm_training_labels:True (harness.py:192)
and label source is human (harness.py:178).",
372         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:33,60",
373         "verdict": "correct",
374         "evidence": "training/gen0/harness.py:236 'no_paid_llm_training_labels':
True; :222 'label_source': 'human (rally-annotator golden CSVs)'. Facts correct; cited lines
(:192, :178) have drifted to :236 and :222.",
375         "suggestedFix": "Update line citations: no_paid_llm_training_labels at
harness.py:236, label_source at :222.",
376         "severity": "low"
377     },
378     {
379         "claim": "cmd_train (__main__.py:231) pulls labels/, joins video<->label by
file MD5 + stem-match labels/<name>.rallies.csv <-> videos/<name>.<ext> (__main__.py:271-280).",
380         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:31,49",
381         "verdict": "correct",
382         "evidence": "backend/worker/__main__.py:332 def cmd_train; :359 lists
{corpus}/labels/; :383-384 fname.replace('.rallies.csv','') stem join. Logic present and
correct; cmd_train line drifted from :231 to :332.",
383         "suggestedFix": "Update cmd_train citation to __main__.py:332.",
384         "severity": "low"
385     },
386     {
387         "claim": "Reuse map: add a /own-model alias in site/server.js:49-51; Worker
env.ASSETS.fetch needs no change (site/src/index.js:84).",
388         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52,73",
389         "verdict": "correct",
390         "evidence": "site/server.js:99 'pathname === \"/own-model\" ? \"/own-
model.html\"' and :41 LOCALIZED_PAGES includes own-model.html (alias shifted off the cited

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:49-51). site/src/index.js:84 'return env.ASSETS.fetch(request)' with no /own-model branch ?
confirms the Worker needs no change.",
391         "suggestedFix": "Update the server.js alias citation (now ~:99, not
:49-51).",
392         "severity": "low"
393     },
394     {
395         "claim": "Reuse precedent: screenshots under site/public/img/ (precedent
site/og.png + make_og.py).",
396         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52",
397         "verdict": "correct",
398         "evidence": "git ls-files: site/make_og.py and site/public/og.png both
exist; site/public/img/ holds annotate-enable.png, annotate-tool.png, studio-*.png. The og asset
path is site/public/og.png (the doc wrote 'site/og.png' loosely), make_og.py is at
site/make_og.py.",
399         "suggestedFix": "Minor: the OG image is at site/public/og.png (not
site/og.png).",
400         "severity": "low"
401     },
402     {
403         "claim": "SPA today: web/src/App.tsx is one static screen, no router
(web/package.json); a correction screen needs react-router-dom.",
404         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,64",
405         "verdict": "stale",
406         "evidence": "grep on web/package.json: no react-router dep (claim 'no
router' still TRUE). But App.tsx is no longer 'one static screen' ? it now imports and
conditionally renders SetupWizard (web/src/App.tsx:11,65,242, P2c first-run wizard from PR #88).
The 'one static screen' framing predates the wizard.",
407         "suggestedFix": "Reword to 'no router; App.tsx renders a first-run
SetupWizard + the ingest panel' so the SPA surface description matches the post-#88 code.",
408         "severity": "low"
409     },
410     {
411         "claim": "SPA correction route depends on a new, unbuilt engine endpoint
PATCH /api/segments/{id} (dead columns
candidate_segments.human_label/notes/confirmed_segment_id); P8 ? Gated.",
412         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,79",
413         "verdict": "correct",
414         "evidence": "grep across backend/ for 'api/segments'/'PATCH' on
candidate_segments finds only the candidate_segments table + query_candidate_segments readers
(database.py:217, eval/*.py); no PATCH /api/segments route exists. P8 is still genuinely unbuilt
as stated.",
415         "suggestedFix": "",
416         "severity": "low"
417     },
418     {
419         "claim": "Relation note: peer of LOCAL_APPLIANCE_ARCHITECTURE.md (the latter
lands with PR #13).",
420         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:6,84",
421         "verdict": "stale",
422         "evidence": "git ls-files confirms docs/LOCAL_APPLIANCE_ARCHITECTURE.md
already exists in the repo; it has long since landed, so the forward-looking 'lands with PR #13'
/ 'cross-link once PR #13 merges' phrasing is outdated.",
423         "suggestedFix": "Drop the 'lands with PR #13' future-tense framing ? the doc
exists now; add the cross-link.",
424         "severity": "low"
425     },
426     {
427         "claim": "rally-annotator is a single-file VLC Lua extension, VERSION 1.6.4,
MIT-licensed, public at github.com/Khelsutra/rally-annotator.",
428         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:29",
429         "verdict": "correct",
430         "evidence": "rally-annotator/vlc/rally_annotator.lua:1 '(v1.6.4)'; LICENSE
first line 'MIT License'; own-model.html:109/191 links to https://github.com/Khelsutra/rally-
annotator (org-link migration was PR #23, captured in the doc's own history).",

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431         "suggestedFix": "",
432         "severity": "low"
433     },
434     {
435         "claim": "$7 tracker: P1/P2/P3/P4 all ? via '(this PR)'.",
436         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:72-75",
437         "verdict": "correct",
438         "evidence": "own-model.html exists (P1); /own-model alias in server.js:99
(P2); nav cross-link index.html:255 + footer :620 (P3); verbatim annotator content + GitHub link
in own-model.html:108-191 (P4). All shipped ? git history shows these in PR #22 (b1110e1,
'annotate-train P1-P4'), so '(this PR)' should now read '#22'.",
439         "suggestedFix": "Replace the '(this PR)' placeholders in the PR column with
#22 (and #23 for the org-link follow-up) so the tracker cites real merged PRs.",
440         "severity": "medium"
441     }
442 ]
443 },
444 {
445     "doc": "docs/DRY_RUN_CUJ.md",
446     "overallStatus": "minor-staleness",
447     "summary": "The doc's substantive engineering claims are accurate and well-
grounded: the SPA back-half + B-P2 front-half ingest screen exist
(web/src/components/IngestPanel.tsx, web/src/hooks/useIngestJob.ts, #56), the engine bucket-
ingest routing is verified in code, and the three cited engine PRs (#280/#281/#289) plus all
four cited engine tests exist. The staleness is in metadata/framing: the header still says
Status: DRAFT though docs/README.md correctly calls it DELIVERED with DoD $6 boxes 1-3 done, and
the doc repeatedly describes the multilingual workflow as \"en/kn/hi\" (3 langs) when the SPA
now supports 6 (es/da/id added in #87). It is a completed tracker and an archive candidate once
the parked droplet box is closed.",
448     "findings": [
449         {
450             "claim": "Status: DRAFT ? owner-gated . ... Goal for this session.",
451             "docLocation": "docs/DRY_RUN_CUJ.md:3",
452             "verdict": "stale",
453             "evidence": "docs/README.md:4 marks this doc '? DELIVERED (2026-06-21)'; DoD
$6 boxes 1-3 are all checked [x] (DRY_RUN_CUJ.md:47-49) and the work shipped (#53/#55/#56/#57).
A DRAFT/'goal for this session' header no longer matches the completed, delivered state.",
454             "suggestedFix": "Flip the header to Status: DONE / DELIVERED (2026-06-21)
and drop 'Goal for this session'; note the only open box (droplet, $6 box 4) is owner-parked.
Consider archiving once that box closes.",
455             "severity": "medium"
456         },
457         {
458             "claim": "The SPA is 'Fully en/kn/hi (the switcher / ?lang=)' and the
multilingual workflow is en/kn/hi.",
459             "docLocation": "docs/DRY_RUN_CUJ.md:11",
460             "verdict": "stale",
461             "evidence": "web/src/i18n/index.ts:25 ? `export const SUPPORTED_LOCALES =
[\"en\", \"kn\", \"hi\", \"es\", \"da\", \"id\"]`; es/da/id were added to the SPA in PR #87
(cf59fbf 'add es/da/id to the marketing site (parity with the SPA's 6 languages)'). The SPA now
supports 6 languages, not 3.",
462             "suggestedFix": "Update 'en/kn/hi' to 'en/kn/hi/es/da/id (6 languages)'
where it describes the current switcher; the Hindi/Kannada run results in $7 stay accurate as
historical record.",
463             "severity": "low"
464         },
465         {
466             "claim": "This SPA front-half (ingest/process UI) is the *only* net-new work
for Path B (= B-P2); the SPA has no ingest/process UI today.",
467             "docLocation": "docs/DRY_RUN_CUJ.md:14",
468             "verdict": "correct",
469             "evidence": "B-P2 was net-new work at draft time and has since shipped:
web/src/components/IngestPanel.tsx + web/src/hooks/useIngestJob.ts (first landed in PR #56
'feat(web): B-P2 ? in-UI bucket ingest', 4525c5d, 2026-06-21). useIngestJob.ts drives POST
/api/process (processVideo) ? poll GET /api/jobs/{id} (getJob) ? hand video_id to App, exactly

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as scoped.",
470         "suggestedFix": "",
471         "severity": "low"
472     },
473     {
474         "claim": "POST /api/process accepts gs://gcs://s3://local:// URIs and skips
the local-existence check for non-local backends; resolve via storage.resolve_input_ref;
gs://?gcs normalization.",
475         "docLocation": "docs/DRY_RUN_CUJ.md:15,68-72",
476         "verdict": "correct",
477         "evidence": "Engine backend/main.py:581 `input_ref =
storage.resolve_input_ref(req.video_path, ...)` and main.py:590 `if input_ref.backend ==
`\"local\" and not os.path.exists(input_ref.key):` (the non-local skip), with the fetch at
main.py:458 `storage.acquire_local_input(...)`; backend/storage/ref.py:60 `if scheme == \"gs\":
scheme = \"gcs\"`. Substance matches; the exact line numbers 159/274/283 cited in the doc have
drifted as the engine shipped further PRs.",
478         "suggestedFix": "Optionally refresh the cited line numbers (now
~main.py:581/590/458, ref.py:60) since the engine surface moved; the routing logic itself is
intact.",
479         "severity": "low"
480     },
481     {
482         "claim": "Tested by
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches,
::test_process_hosted_mode_rejects_nonlocal_uri, ::test_process_unknown_scheme_returns_400,
::test_cli_nonlocal_input_fetches_and_keeps_uri.",
483         "docLocation": "docs/DRY_RUN_CUJ.md:15,76",
484         "verdict": "correct",
485         "evidence": "Engine repo grep: tests/test_api_endpoints.py:508
test_process_nonlocal_uri_skips_exists_and_fetches; :577
test_process_unknown_scheme_returns_400; :608 test_process_hosted_mode_rejects_nonlocal_uri;
:627 test_cli_nonlocal_input_fetches_and_keeps_uri. All four exist with the asserted names.",
486         "suggestedFix": "",
487         "severity": "low"
488     },
489     {
490         "claim": "B-P1 (gs://?/api/process?API-DB video_id) already shipped in
engine M2 / #280; the job-drain hang fixed in #281; the compile-from-bucket 500 fixed in #289.",
491         "docLocation": "docs/DRY_RUN_CUJ.md:15,63,68,88",
492         "verdict": "correct",
493         "evidence": "Engine git log: e111add 'feat(serving): M2 ? URI-aware input
... (#280)'; 41d45fd 'fix(server): drain async jobs under `python -m backend.main` (double-
module hang) (#281)'; e68b583 'fix(api): resolve a bucket-ingested video's source to a local
file before stitching (#289)'. All three PRs exist with descriptions matching the doc.",
494         "suggestedFix": "",
495         "severity": "low"
496     },
497     {
498         "claim": "DoD $6 box 3: in-UI Path-B bucket?reel CUJ DONE (live e2e
2026-06-21) ? 12 rallies loaded ? reel downloaded; box 4 (CPU droplet) still open/un-
provisioned.",
499         "docLocation": "docs/DRY_RUN_CUJ.md:49-50",
500         "verdict": "correct",
501         "evidence": "Box 3 is corroborated by PR #57 'docs(dry-run): Path-B live e2e
DONE ? in-UI bucket?reel CUJ + screencast (DoD $6 box 3 ?)' and the merged B-P2 SPA (#56).
docs/README.md:6-7 confirms box 4 (droplet, option 3) remains 'parked by the owner (GPU
subscription)'. The run-result specifics (12 rallies, reel sizes, GCS screencast paths) are
runtime artifacts not locally reproducible.",
502         "suggestedFix": "",
503         "severity": "low"
504     },
505     {
506         "claim": "Run-result specifics: 12 rallies / video_id 05e0e577 /
50.9MB+76.5s reels / screencasts at gs://khelsutra-rally-corpus/promo/*.mp4 and the staged
bucket video gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4.",

```

```

507         "docLocation": "docs/DRY_RUN_CUJ.md:54-60,82-85",
508         "verdict": "unverifiable",
509         "evidence": "These are GCS bucket objects and a one-off local run's outputs;
they cannot be checked from the repo. No counter-evidence found ? the surrounding code/PR claims
that frame them are all verified.",
510         "suggestedFix": "",
511         "severity": "low"
512     }
513 ]
514 },
515 {
516     "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
517     "overallStatus": "significant-staleness",
518     "summary": "A still-active DRAFT tracker (last updated 2026-06-21) whose $9
progress and several \"Absent (net-new)\" engine-fact anchors have drifted: the engine has since
shipped /api/capabilities, exposure-safety boot posture + an edge-auth layer, and merged
native_wasb_runner to master, while the alpha deploy went LIVE on 2026-06-22
(app.khelsutra.guru). The khelsutra-side claims (PR refs #11/#56/#64/#66, IngestPanel,
VITE_API_BASE) check out; the staleness is concentrated in engine status and the deploy/DoD
progress. Not yet an archive candidate (open owner gates remain) but needs a refresh pass.",
519     "findings": [
520     {
521         "claim": "$3 lists `GET /api/capabilities` under 'Absent (net-new)': ...
`[verified ? grep returns none]`; $4.6 Reuse map lists it under 'Net-new'; $9 T7 (capabilities
+ health + banner) is ? Todo.",
522         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:49,113,164",
523         "verdict": "stale",
524         "evidence": "engine origin/master backend/main.py:340
`@app.get(`/api/capabilities`)` (and main.py:307 `/api/health`, main.py:334 `/api/version`)
all exist on master. git grep `/api/capabilities` origin/master returns a hit.",
525         "suggestedFix": "Move `/api/capabilities` out of 'Absent (net-new)' and out
of $4.6 'Net-new'; mark the engine half of T7 (capabilities+health) as shipped (??), leaving
only the SPA banner pending.",
526         "severity": "high"
527     },
528     {
529         "claim": "TL;DR + Risk table: 'the engine has zero app-layer auth and CORS
`*` `[verified backend/main.py:42]` ? structural safety only because the engine binds
127.0.0.1.",
530         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:14,30,103,128",
531         "verdict": "stale",
532         "evidence": "engine origin/master backend/main.py:82 uses
`allow_origins=cors_origins_from_env()` (default `['*']` but env-lockable) with
`allow_credentials=False`; main.py:109 adds TrustedHostMiddleware (DNS-rebinding/Host
allowlist); backend/api/auth.py + config/startup.py enforce_startup_checks + M9 `Cf-Access-
Authenticated-User-Email`/`owner_id` now exist on master. CORS is no longer a hardcoded bare
`*`, and an app-layer edge-auth integration now exists.",
533         "suggestedFix": "Reword to: CORS default is `*` but now env-configurable
(allow_credentials off), and the engine now ships a Host-header guard + an edge-auth integration
(Cf-Access trust) + exposure-safety boot posture ? so 'zero app-layer auth' is no longer
literally true. Re-anchor the line (main.py:42 no longer points at CORS).",
534         "severity": "high"
535     },
536     {
537         "claim": "$3 / $4.3: 'Native WASB is in-flight on branch `feat/native-wasb-
runner`: `native_wasb_runner.py` exists `[verified]`'.",
538         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:45,96",
539         "verdict": "stale",
540         "evidence": "engine origin/master now contains
backend/pipeline/detectors/native_wasb_runner.py and tests/test_native_wasb_runner.py (git ls-
tree origin/master). The runner has merged to master; the `feat/native-wasb-runner` branch still
exists but the file is no longer branch-only. (trajectory_hybrid._build_native_runner still
raises NotImplementedError at trajectory_hybrid.py:61 ? that sub-claim is still correct.)",
541         "suggestedFix": "Change 'in-flight on branch feat/native-wasb-runner' to
'native_wasb_runner.py is on master; trajectory_hybrid still refuses, fusion builds it; owner

```



```

byte-parity gate still pending.'"
542         "severity": "medium"
543     },
544     {
545         "claim": "$9 alpha-deploy block: 'the actual stand-up (tunnel + Pages + CF
Access + key rotation) is the **owner** step'; $11 DoD box 1 (CUJ-1 e2e on droplet behind gate)
and box 2 (raw :8000 refused, tested) unchecked.",
546         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:175,182,183",
547         "verdict": "stale",
548         "evidence": "deploy/droplet/README.md:100-106 '? Live-verified on the
droplet (2026-06-22)': engine booted on 168.144.126.176; suite 1313 passed; boot-posture fail-
fast, Host-guard 400?200, path-jail, T4 upload?jail, and edge-auth 401-gate all exercised live;
full DNS?Cloudflare?Access?tunnel?engine chain confirmed; app.khelsutra.guru up. Only Gemini-key
rotation + a browser OTP login remain.",
549         "suggestedFix": "Update $9 to note the alpha stand-up is LIVE (2026-06-22,
app.khelsutra.guru, posture test exercised) with only key rotation + browser-OTP login
outstanding; tick/qualify the corresponding DoD boxes.",
550         "severity": "high"
551     },
552     {
553         "claim": "Status header: 'Status: DRAFT ... Last updated: 2026-06-21' with
$9 the source of truth.",
554         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:3",
555         "verdict": "stale",
556         "evidence": "khelsutra git log shows merges through #94 (master HEAD
50b2bd7) and the alpha went live 2026-06-22, both after the doc's 2026-06-21 'last updated'.
Multiple $9 rows and engine facts have moved since (T7 engine half, alpha deploy, native_wasb on
master).",
557         "suggestedFix": "Bump 'Last updated' and consider promoting DRAFT?IN
PROGRESS; refresh $9 rows to current reality.",
558         "severity": "medium"
559     },
560     {
561         "claim": "D4 'Embed RallyPicker ... PR [#11]' and $9 T4 '[$66]', T12 'SPA
#56', alpha-deploy '[$64]' / 'VITE_API_BASE'.",
562         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:33,161,169,175",
563         "verdict": "correct",
564         "evidence": "khelsutra git log: cb74ce5 '...RallyPicker (#11)'; ca619fb 'T4
? in-UI file upload (#66)'; 4525c5d 'B-P2 ? in-UI bucket ingest (#56)'; 8bbc557 'VITE_API_BASE
(#64)'. web/src/components/IngestPanel.tsx handles both chunked upload (T4) and a gs:// URI
(T12); web/src/api/client.ts:25 reads VITE_API_BASE.",
565         "suggestedFix": "",
566         "severity": "low"
567     },
568     {
569         "claim": "T12 (bucket-URI ingest) marked ?: 'engine accepts URIs (M2/#280) +
SPA ingest screen shipped (B-P2, #56); in-UI gs:///reel CUJ proven live e2e (DRY_RUN_CUJ.md
$9)'.",
570         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:169",
571         "verdict": "correct",
572         "evidence": "docs/DRY_RUN_CUJ.md:80-92 $9 records the live in-UI gs:///reel
CUJ (12 rallies ? reel downloaded) and the compile-from-bucket 500 fix (engine #289);
IngestPanel.tsx accepts a gs:// URI. khelsutra #56 (B-P2) is merged.",
573         "suggestedFix": "",
574         "severity": "low"
575     },
576     {
577         "claim": "T11 retention marked ?: sweep tool shipped (engine #299), auto-
scheduling tracked (engine #301).",
578         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:168,130",
579         "verdict": "unverifiable",
580         "evidence": "These reference the engine repo's GitHub PR #299 / issue #301
(Khelsutra/badminton-highlight-indexer). Not falsifiable from the local khelsutra checkout; the
local engine checkout is on a feature branch and I did not confirm the specific sweep tool by PR
number. The khelsutra-side row metadata is internally consistent (#69 flipped T11 to ? per git

```

```

log bea8a2a).",
581         "suggestedFix": "",
582         "severity": "low"
583     },
584     {
585         "claim": "$3 Droplet reality: co-hosts the marketing site at
`/srv/khelsutra-site`, node `:8787`, engine at `~/rally`; MinIO 9000/9001 + GCS emulator 9023.",
586         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48",
587         "verdict": "unverifiable",
588         "evidence": "These are live-droplet filesystem/port facts (168.144.126.176)
not checkable from the local repo. deploy/DEPLOY_ALPHA.md:31 and deploy/droplet/README.md:4
corroborate the IP/CPU-no-GPU/blr1 framing, but the exact paths/ports (/srv/khelsutra-site,
:8787, ~/rally) are not in-repo verifiable.",
589         "suggestedFix": "",
590         "severity": "low"
591     },

```

5. assistant (2026-06-22T09:46:32.661Z)

Now let me extract the PORTABILITY.md findings from the audit. The truncated view only showed up to line 591. Let me search for the PORTABILITY.md entries in the JSON.

6. user (2026-06-22T09:46:34.974Z)

```

55:         "claim": "Docs: docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md.",
58:         "evidence": "git ls-files: both docs/REQUEST_ACCESS_FORM.md and
docs/PORTABILITY.md are tracked.",
793:         "doc": "docs/PORTABILITY.md",
799:         "docLocation": "docs/PORTABILITY.md:8-9",
807:         "docLocation": "docs/PORTABILITY.md:10-11",
815:         "docLocation": "docs/PORTABILITY.md:12-13",
823:         "docLocation": "docs/PORTABILITY.md:16-23",
831:         "docLocation": "docs/PORTABILITY.md:25",
839:         "docLocation": "docs/PORTABILITY.md:25-26",
847:         "docLocation": "docs/PORTABILITY.md:28-29",
855:         "docLocation": "docs/PORTABILITY.md:31-32",
863:         "docLocation": "docs/PORTABILITY.md:39-42",
871:         "docLocation": "docs/PORTABILITY.md:47-55",
879:         "docLocation": "docs/PORTABILITY.md:64",
887:         "docLocation": "docs/PORTABILITY.md:57-61",
895:         "docLocation": "docs/PORTABILITY.md:79",
968:         "evidence": "docs/PORTABILITY.md:1-3 'Portability ? Cloudflare now, always-on VM
if needed'. site/server.js:8,11,13,124,138 implements a Node http server using node:sqlite
(DatabaseSync) and the shared src/store.js ? the Node+SQLite VM host described.",
1165:         "evidence": "site/server.js:1-14 is an 'Always-on VM host for the KhelSutra
site ... the portability fallback if Cloudflare acts out ? see docs/PORTABILITY.md', sharing the
same handleRequest from src/index.js. docs/PORTABILITY.md exists. The site test run shows a full
'[VM host]' suite (POST persists to SQLite, serves /own-model, flyers, locales). The README has
zero mention of server.js, the VM host, the droplet, or PORTABILITY.md.",
1166:         "suggestedFix": "Add a short section: the same handler runs in two places ? the
Cloudflare Worker (src/index.js, this README's focus) and an always-on VM host (server.js, the
portability fallback; see docs/PORTABILITY.md), used by the alpha droplet.",
1577:         "doc": "docs/PORTABILITY.md",
1579:         "docLocation": "docs/PORTABILITY.md:25",
1586:         "doc": "docs/PORTABILITY.md",
1588:         "docLocation": "docs/PORTABILITY.md:64",
1662:         "evidence": "site/server.js:1-14 is an 'Always-on VM host for the KhelSutra site
... the portability fallback if Cloudflare acts out ? see docs/PORTABILITY.md', sharing the same
handleRequest from src/index.js. docs/PORTABILITY.md exists. The site test run shows a full '[VM
host]' suite (POST persists to SQLite, serves /own-model, flyers, locales). The README has zero
mention of server.js, the VM host, the droplet, or PORTABILITY.md.",
1663:         "suggestedFix": "Add a short section: the same handler runs in two places ? the
Cloudflare Worker (src/index.js, this README's focus) and an always-on VM host (server.js, the
portability fallback; see docs/PORTABILITY.md), used by the alpha droplet.",

```

7. assistant (2026-06-22T09:46:37.777Z)

The PORTABILITY.md perDoc block is at lines 793-900, and staleFindings at 1577-1600. Let me read both ranges.

8. user (2026-06-22T09:46:40.685Z)

```
793         "doc": "docs/PORTABILITY.md",
794         "overallStatus": "minor-staleness",
795         "summary": "PORTABILITY.md is overwhelmingly accurate and code-backed: every
cited file (site/src/index.js, src/store.js, server.js, schema.sql, the test files, and the
provision/teardown/smoke scripts) exists, the described adapters/helpers/routes match the code,
and the full site suite (86 tests, both Cloudflare-D1 and VM node:sqlite paths) is green. Only
two minor blemishes: an internal Node-version inconsistency (runbook says \"Node 22+\" while the
provisioning section, CI, and scripts all require Node >=24 for unflagged node:sqlite) and a
slightly imprecise test command/handler signature. Not an archive candidate ? it documents live,
in-use portability infrastructure.",
796         "findings": [
797             {
798                 "claim": "Logic is runtime-agnostic ? src/index.js exports pure helpers
validate, toRow, isBot, handleRequest with no host-specific calls.",
799                 "docLocation": "docs/PORTABILITY.md:8-9",
800                 "verdict": "correct",
801                 "evidence": "site/src/index.js:20 (isBot), :24 (validate), :31 (toRow), :55
(handleRequest) ? all exported; uses Web-standard Request/Response (json() builds new Response,
handleRequest reads request.json()).",
802                 "suggestedFix": "",
803                 "severity": "low"
804             },
805             {
806                 "claim": "All DB coupling is in src/store.js: d1Store (Cloudflare D1) and
sqliteStore (node:sqlite / better-sqlite3); handleRequest(request, store) takes the store by
injection.",
807                 "docLocation": "docs/PORTABILITY.md:10-11",
808                 "verdict": "correct",
809                 "evidence": "site/src/store.js:20 (d1Store), :29 (sqliteStore), shared
INSERT_SQL:9. Injection confirmed. Note signature is now handleRequest(request, store, notify) ?
3rd param added (site/src/index.js:55); doc's 2-arg form is slightly stale but the injection
point it describes is accurate.",
810                 "suggestedFix": "Optionally update to handleRequest(request, store, notify)
to reflect the added notify injection.",
811                 "severity": "low"
812             },
813             {
814                 "claim": "One schema.sql + plain-SQLite INSERT, byte-for-byte compatible
between D1 and a VM SQLite file.",
815                 "docLocation": "docs/PORTABILITY.md:12-13",
816                 "verdict": "correct",
817                 "evidence": "site/schema.sql:3 CREATE TABLE IF NOT EXISTS access_requests
(plain SQLite); site/src/store.js:9 INSERT_SQL is dialect-neutral and used by both stores;
server.js:22 execs the same schema.sql.",
818                 "suggestedFix": "",
819                 "severity": "low"
820             },
821             {
822                 "claim": "Two-hosts table: CF entry src/index.js (Worker fetch) / VM entry
server.js (node:http); CF static env.ASSETS / VM server.js serves public/; storage
d1Store(env.DB) vs sqliteStore(db); country cf-ipcountry header.",
823                 "docLocation": "docs/PORTABILITY.md:16-23",
824                 "verdict": "correct",
825                 "evidence": "src/index.js:77-85 (default fetch, env.ASSETS.fetch fallback);
server.js:8 createServer/node:http, :18 PUBLIC=public, :96 serveStatic; src/index.js:67
request.headers.get('cf-ipcountry'); store.js d1Store/sqliteStore.",
826                 "suggestedFix": "",
827                 "severity": "low"
```

```

828     },
829     {
830         "claim": "`node --test src/` runs BOTH paths.",
831         "docLocation": "docs/PORTABILITY.md:25",
832         "verdict": "stale",
833         "evidence": "package.json `test` (no src/ arg) is what CI
runs (.github/workflows/test.yml:25 `npm test`). The split targets are test:cf / test:vm. `node
--test src/` is not the command CI uses; the bare `node --test` discovers and runs all
*.test.js.",
834         "suggestedFix": "Change to: `npm test` (node --test) runs BOTH paths, and CI
runs it on every PR.",
835         "severity": "low"
836     },
837     {
838         "claim": "CI (.github/workflows/test.yml, Node 24) runs both paths on every
PR.",
839         "docLocation": "docs/PORTABILITY.md:25-26",
840         "verdict": "correct",
841         "evidence": ".github/workflows/test.yml:2-6 triggers on push:[master] +
pull_request; :17 node-version `24`; :25 runs `npm test` (all paths). Ran `node --test`
locally: 86 tests pass, 0 fail, including [VM host] booted-server.js round-trip and the
Worker/D1 logic.",
842         "suggestedFix": "",
843         "severity": "low"
844     },
845     {
846         "claim": "src/server.test.js boots the real server.js (node:http +
node:sqlite) and round-trips it over HTTP into SQLite.",
847         "docLocation": "docs/PORTABILITY.md:28-29",
848         "verdict": "correct",
849         "evidence": "site/src/server.test.js:36 spawn(process.execPath,
[serverPath]) boots the real server.js with PORT/DB_PATH; :82-98 POSTs /api/request-access then
reads the row back from the SQLite file via DatabaseSync. Test passes.",
850         "suggestedFix": "",
851         "severity": "low"
852     },
853     {
854         "claim": "For a live check: SSH_TARGET=root@host scripts/droplet-smoke.sh
(scp + boot + round-trip + cleanup).",
855         "docLocation": "docs/PORTABILITY.md:31-32",
856         "verdict": "correct",
857         "evidence": "site/scripts/droplet-smoke.sh exists; header documents
SSH_TARGET=root@1.2.3.4 usage, scp site/ to droplet, boot server.js, round-trip, cleanup temp
DB. bash -n parses clean (CI shell parse-check step).",
858         "suggestedFix": "",
859         "severity": "low"
860     },
861     {
862         "claim": "scripts/provision-vm.sh pushes site/ via git archive (private-
repo-safe), installs Node >=24, installs+starts a systemd service, opens the firewall, and
verifies it serves; idempotent.",
863         "docLocation": "docs/PORTABILITY.md:39-42",
864         "verdict": "correct",
865         "evidence": "site/scripts/provision-vm.sh header + body: 'pushes site/ via
git archive (PRIVATE-REPO-SAFE)', NODE_MAJOR default 24 with 'must be >= 24 so node:sqlite is
unflagged', systemd service, firewall, VERIFIES serving, 'Idempotent'.",
866         "suggestedFix": "",
867         "severity": "low"
868     },
869     {
870         "claim": "scripts/teardown-vm.sh exports the SQLite DB to a local
timestamped backup BEFORE destroying, stops the service, and optionally destroys the DO droplet
(DESTROY=1 DROPLET_ID=<id> via doctl); RESTORE_DB resume flow.",
871         "docLocation": "docs/PORTABILITY.md:47-55",
872         "verdict": "correct",

```

```

873         "evidence": "site/scripts/teardown-vm.sh: STAMP+OUT timestamped backup,
scp's access.db out BEFORE destroy, systemctl stop khelsutra-site, `if [ DESTROY=1 ] && [
DROPLET_ID ]` block. provision-vm.sh accepts RESTORE_DB to scp the backup in before start.",
874         "suggestedFix": "",
875         "severity": "low"
876     },
877     {
878         "claim": "VM deploy runbook: Node 22+ only ? no npm install (node:sqlite is
built in).",
879         "docLocation": "docs/PORTABILITY.md:64",
880         "verdict": "stale",
881         "evidence": "Internally inconsistent with the doc's own provisioning section
(line 40: Node >=24 'so node:sqlite is unflagged') and the repo: provision-vm.sh NODE_MAJOR
default 24 + 'must be >= 24', .github/workflows/test.yml:17 node-version 24, server.js:1 also
says 'Node 22+'. node:sqlite was flagged/experimental before Node ~22.5 and only stabilized
later; a reader running the runbook on Node 22 may hit a flagged/unavailable node:sqlite. The
authoritative requirement in code is >=24.",
882         "suggestedFix": "Change 'Node 22+ only' to 'Node 24+ only (node:sqlite is
unflagged at >=24)' in both the runbook (line 64) and server.js:1, to match provision-vm.sh /
CI.",
883         "severity": "medium"
884     },
885     {
886         "claim": "Verified live (2026-06-20): a fresh DO Ubuntu-24.04 droplet went
bare -> operational via provision-vm.sh; teardown exported a 12 KB DB; re-provision with
RESTORE_DB preserved rows:2; droplet then destroyed.",
887         "docLocation": "docs/PORTABILITY.md:57-61",
888         "verdict": "unverifiable",
889         "evidence": "This is a historical live-run claim against an external DO
droplet; it cannot be confirmed or refuted from the repo. The supporting scripts
(provision/teardown with export+RESTORE_DB) exist and bash -n parse, and the booted-server
round-trip is proven in CI, so the claim is consistent with shipped code, but the live event
itself is not falsifiable here.",
890         "suggestedFix": "",
891         "severity": "low"
892     },
893     {
894         "claim": "server.js is intentionally dependency-free.",
895         "docLocation": "docs/PORTABILITY.md:79",
896         "verdict": "correct",
897         "evidence": "site/server.js:8-14 imports only node: builtins (node:http,
node:fs/promises, node:path, node:sqlite) plus local ./src modules; package.json has no runtime
deps (jsdom + playwright are devDependencies). server.js:1 'dependency-free (Node 22+ only)'.",
898         "suggestedFix": "",
899         "severity": "low"
900     }
901 ]
902 },
903 {
904     "doc": "docs/README.md",
905     "overallStatus": "minor-staleness",
906     "summary": "The index is mostly accurate and freshly maintained: the
DRY_RUN_CUJ, archived FRIENDLY_AND_MULTILINGUAL, LOCAL_APPLIANCE, PER_RALLY_SELECTION,
REQUEST_ACCESS_FORM, PORTABILITY, and engine-planning-docs entries all verify against code/git.
Two real defects: the MULTILINGUAL_EXPANSION entry is stale (shown as a not-yet-started
DRAFT/[#58] proposal but the doc is IN PROGRESS with Phase A es/da/id already shipped in #67),
and a tracked doc (ANNOTATE_AND_TRAIN.md) is missing from the index entirely.",
907     "findings": [
908     {
909         "claim": "MULTILINGUAL_EXPANSION.md is a 'proposal to add 7 more locales
(th/id/da/es/zh-Hans/ja/ko) ... DRAFT ? owner-gated (approval to start), [#58].'",
910         "docLocation": "docs/README.md:16-18",
911         "verdict": "stale",
912         "evidence": "docs/MULTILINGUAL_EXPANSION.md:3 status header reads 'IN
PROGRESS ? Phase A (es/da/id) shipped (#67); Phases B (Thai) / C (CJK) owner-gated'. Git

```

confirms PR #67 'feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs'. site/public/locales/ now contains es, da, id directories and web/src/i18n/index.ts:25 lists SUPPORTED_LOCALES = en, kn, hi, es, da, id. So es/da/id are no longer 'to add', and the doc is not a not-yet-started DRAFT.",

9. user (2026-06-22T09:46:41.885Z)

```
1573         "suggestedFix": "Reference it as docs/REQUEST_ACCESS_FORM.md.",
1574         "severity": "low"
1575     },
1576     {
1577         "doc": "docs/PORTABILITY.md",
1578         "claim": "`node --test src/` runs BOTH paths.",
1579         "docLocation": "docs/PORTABILITY.md:25",
1580         "verdict": "stale",
1581         "evidence": "package.json `\"test\": \"node --test\"` (no src/ arg) is what CI
runs (.github/workflows/test.yml:25 `npm test`). The split targets are test:cf / test:vm. `node
--test src/` is not the command CI uses; the bare `node --test` discovers and runs all
*.test.js.",
1582         "suggestedFix": "Change to: `npm test` (node --test) runs BOTH paths, and CI
runs it on every PR.",
1583         "severity": "low"
1584     },
1585     {
1586         "doc": "docs/PORTABILITY.md",
1587         "claim": "VM deploy runbook: Node 22+ only ? no npm install (node:sqlite is
built in).",
1588         "docLocation": "docs/PORTABILITY.md:64",
1589         "verdict": "stale",
1590         "evidence": "Internally inconsistent with the doc's own provisioning section
(line 40: Node >=24 'so node:sqlite is unflagged') and the repo: provision-vm.sh NODE_MAJOR
default 24 + 'must be >= 24', .github/workflows/test.yml:17 node-version 24, server.js:1 also
says 'Node 22+'. node:sqlite was flagged/experimental before Node ~22.5 and only stabilized
later; a reader running the runbook on Node 22 may hit a flagged/unavailable node:sqlite. The
authoritative requirement in code is >=24.",
1591         "suggestedFix": "Change 'Node 22+ only' to 'Node 24+ only (node:sqlite is
unflagged at >=24)' in both the runbook (line 64) and server.js:1, to match provision-vm.sh /
CI.",
1592         "severity": "medium"
1593     },
1594     {
1595         "doc": "docs/README.md",
1596         "claim": "MULTILINGUAL_EXPANSION.md is a 'proposal to add 7 more locales
(th/id/da/es/zh-Hans/ja/ko) ... DRAFT ? owner-gated (approval to start), [#58].'",
1597         "docLocation": "docs/README.md:16-18",
1598         "verdict": "stale",
1599         "evidence": "docs/MULTILINGUAL_EXPANSION.md:3 status header reads 'IN PROGRESS ?
Phase A (es/da/id) shipped (#67); Phases B (Thai) / C (CJK) owner-gated'. Git confirms PR #67
'feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs'. site/public/locales/
now contains es, da, id directories and web/src/i18n/index.ts:25 lists SUPPORTED_LOCALES =
en, kn, hi, es, da, id. So es/da/id are no longer 'to add', and the doc is not a not-yet-started
DRAFT.",
1600         "suggestedFix": "Update to: '*`MULTILINGUAL_EXPANSION.md`* ? `IN PROGRESS`:'
Phase A (es/da/id) shipped ([#67]); Thai (B) / CJK (C) remain owner-gated. Adds locales beyond
the en/kn/hi foundation; peer of the archived `FRIENDLY_AND_MULTILINGUAL.md`.'",
1601         "severity": "high"
1602     },
```

10. assistant (2026-06-22T09:46:48.234Z)

I now have my worklist. There are 2 stale findings for docs/PORTABILITY.md:

1. **PORTABILITY.md:25** ? `node --test src/` claim is stale; should be `npm test` (node --test).
2. **PORTABILITY.md:64** ? "Node 22+ only" should be "Node 24+ only". (The doc half; the

suggestedFix also mentions `server.js:1` which is a CODE change ? record under codeChangesNeeded, do not edit.)

There are also two low-severity "optional" findings in perDoc (not in staleFindings, marked verdict=correct with optional suggestions): the `handleRequest(request, store, notify)` 3-arg signature (PORTABILITY.md:11) and the README cross-reference. These are verdict=correct (not stale), so I should NOT act on them per the instructions (only act on stale findings; skim unverifiable but only act if positively verifiable). I'll skip the optional "correct" ones.

Let me re-verify the two stale findings against the live khelsutra repo before editing.

11. user (2026-06-22T09:46:51.389Z)

```
=== package.json scripts ===
  "scripts": {
    "test": "node --test",
    "test:cf": "node --test src/index.test.js",
    "test:vm": "node --test src/server.test.js",
    "test:deploy": "node --test src/deploy-config.test.js src/assets.test.js",
    "test:layout": "node --test src/layout.browser.mjs",
    "copy:locales": "node scripts/copy-locales.mjs",
    "predeploy": "node scripts/copy-locales.mjs",
    "deploy": "wrangler deploy",
=== test.yml relevant lines ===
17:      node-version: "24" # node:sqlite is unflagged; server.test.js boots the VM host
21:      # script in a DOM); the Cloudflare Worker + VM-host RUNTIMES stay dependency-free. `npm
test`
25:      - run: npm test
43:      node-version: "24"
64:      node-version: "24"
=== provision-vm NODE_MAJOR ===
10:# [REMOTE_DIR=/srv/khelsutra-site] [PORT=8787] [NODE_MAJOR=24] [RESTORE_DB=./backup.db] \
13:# NODE_MAJOR must be >= 24 so `node:sqlite` is unflagged (server.js imports it directly).
19:NODE_MAJOR="${NODE_MAJOR:-24}"
40:echo "? [3/5] remote setup (Node $NODE_MAJOR + systemd unit + firewall) ..."
41:ssh "${SSH0[@]}" "$SSH_TARGET" "NODE_MAJOR='$NODE_MAJOR' DIR='$DIR' PORT='$PORT' bash -s"
<<'REMOTE'
46: [ "$(node -p 'process.versions.node.split('.')[0]')" -ge "$NODE_MAJOR" ] && have_node=1
49: echo " installing Node $NODE_MAJOR (NodeSource) ..."
52: curl -fsSL "https://deb.nodesource.com/setup_${NODE_MAJOR}.x" | bash - >/dev/null
=== server.js header ===
// Always-on VM host for the KhelSutra site ? dependency-free (Node 22+ only).
// Serves public/ and POST /api/request-access into SQLite (node:sqlite), using the
// SAME handler logic as the Cloudflare Worker (src/index.js). This is the portability
```

12. assistant (2026-06-22T09:46:59.279Z)

Both findings confirmed:

1. **PORTABILITY.md:25** ? `package.json` "test" script is `node --test` (no `src/` arg), and CI runs `npm test` (test.yml:25). The doc's `node --test src/` is wrong. Confirmed stale.
2. **PORTABILITY.md:64** ? The repo authoritatively requires Node >=24 (provision-vm.sh:13,19 `NODE\_MAJOR >= 24`; test.yml:17 node-version "24"). The runbook says "Node 22+" ? internally inconsistent. Confirmed stale for the doc half. `server.js:1` also says "Node 22+" but that is a CODE change ? codeChangesNeeded.

Let me verify the exact doc text around lines 25-26 and 64 to make surgical edits. I have the doc content already. Line 25-26 reads:

```
...
**Enforced, not asserted.** `node --test src/` runs BOTH paths and CI
(`.github/workflows/test.yml`,
Node 24) runs them on every PR, so the repo stays compatible with both hosts:
...
```

Line 64 reads:

```

Node 22+ only ? **\*\*no `npm install`\*\*** (node:sqlite is built in).

```

Now applying the two surgical edits.

13. user (2026-06-22T09:47:01.651Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PORTABILITY.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

14. user (2026-06-22T09:47:02.677Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PORTABILITY.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

15. assistant (2026-06-22T09:47:05.519Z)

Both edits applied. Now I'll return the structured report.

16. user (2026-06-22T09:47:21.362Z)

Structured output provided successfully

17. assistant (2026-06-22T09:47:26.474Z)

I've completed the staleness fix for `docs/PORTABILITY.md`. Two stale findings were re-verified against the live khelsutra repo and applied as surgical edits:

1. ****Line 25**** ? `node --test src/` ? `npm test` (`node --test`), matching `site/package.json`'s actual `test` script and the `npm test` CI invocation in `.github/workflows/test.yml:25`.
2. ****Line 64**** ? "Node 22+ only" ? "Node 24+ only (node:sqlite is unflagged at >=24)", matching the repo's authoritative `>=24` requirement in `provision-vm.sh` and CI.

One related code change is recorded but not applied (outside my assigned doc): the matching `Node 22+` comment in `site/server.js:1`. Two optional "correct"/"unverifiable" items were skipped per the act-only-on-stale-findings rule.